MODULE 31 · DOMAIN 5 - NETWORKING AND CONTENT DELIVERY

Route 53

Hosted zones, alias records, routing policies, Resolver, Profiles, DNS Firewall and Global Resolver.

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Route 53 is three products: a **registrar**, an **authoritative DNS service** that answers for your domains, and a **resolver** that asks questions on behalf of your VPCs.

Hosted zones

- A **public hosted zone** answers queries from anywhere on the internet. A **private hosted zone** answers only from **associated VPCs**.
- A private zone can be associated with **many VPCs, across accounts and Regions**, which is how a shared internal namespace works.
- Private zones need **enableDnsSupport and enableDnsHostnames** on the VPC. Module 28 flagged these; they are the usual cause of resolution failing.
- Every hosted zone is created with an **NS** and an **SOA** record. Deleting or editing those breaks delegation.

> ON THE EXAM
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> A private hosted zone is invisible outside its associated VPCs. Queries from anywhere else get no answer at all, not a refusal.

Split-horizon DNS

- The **same domain name** can exist in a public zone and a private zone at once, with different answers.
- Queries from an associated VPC get the **private** answer. Queries from anywhere else get the **public** one.
- So internal traffic reaches an internal load balancer by private address, while external traffic reaches the public one, using one hostname.
- The private zone wins for the whole domain within the VPC, so anything not defined there simply does not resolve.

> ON THE EXAM
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> That last point is the trap. Creating a private zone for example.com means subdomains you did not define stop resolving inside the VPC, even public ones.

Alias or CNAME?

ALIAS RECORD

Route 53's own extension

- Works at the zone apex: example.com itself
- Queries to AWS targets are free
- Follows the target's IP changes automatically
- Supports evaluate target health

CNAME RECORD

Standard DNS

- Cannot exist at the zone apex. This is a DNS rule, not an AWS limit
- Charged per query
- Points at any DNS name, inside or outside AWS
- Adds an extra resolution step

THE TIEBREAKER

"example.com must point at the load balancer" can only be an alias. The DNS standard forbids a CNAME coexisting with the SOA and NS records at the apex.

The routing policies

POLICY	CHOOSES BY
Simple	Nothing. One record, and any values are returned in random order
Weighted	A weight from 0 to 255. Proportional split for testing or gradual shifts
Latency	Lowest measured network latency to an AWS Region
Failover	Primary while healthy, otherwise secondary. Module 14 covered this
Geolocation	Where the user is: continent, country, or US state
Geoproximity	Distance to a location, adjustable with a bias. Needs Traffic Flow
Multivalue answer	Returns up to 8 healthy records at random
IP-based	The CIDR block the query came from

ON THE EXAM

Latency routes by measured network performance; geolocation routes by where the user is. Those are different, and questions exploit the difference.

Weighted, latency and multivalue in detail

- **Weighted**: weights are relative, not percentages. A weight of **0** means never used, unless every record is 0, in which case all are used.
- **Latency** is measured network latency to a Region, which is **not** the same as geographic closeness.
- **Multivalue answer** returns up to eight **healthy** records. Simple routing returns all values whether they are healthy or not.
- Multivalue answer is **not a load balancer**. There is no session awareness, no connection draining and no capacity awareness.

> ON THE EXAM
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> Multivalue answer is health-aware simple routing. Its real use is improving availability for something with no load balancer in front of it.

Geolocation or geoproximity?

GEOLOCATION

Where the user is

- Continent, country, or US state
- For compliance, licensing and language
- Always configure a default record
- No concept of distance or adjustment

GEOPROXIMITY

How far from your resource

- Distance between user and resource location
- A bias from -99 to +99 expands or shrinks a region
- Requires Traffic Flow
- For shifting load between sites gradually

THE TIEBREAKER

Geolocation answers "which country is this user in?". Geoproximity answers "which of my sites is nearest, and how much do I want to skew that?"

Records, TTL and Traffic Flow

- **Evaluate target health** on an alias to a load balancer uses the target's own health, with no separate health check to create or pay for.
- **TTL** is how long resolvers cache an answer. Lower means faster change and more queries; higher means cheaper and slower.
- **Alias records to AWS targets have no configurable TTL.** Route 53 manages it.
- **Traffic Flow** is a visual editor producing a versioned **traffic policy**, combining several routing policies into one tree.
- Traffic policies are **versioned and can be rolled back**, which ordinary record edits cannot.

> ON THE EXAM
>
> Lower the TTL days before a planned migration, not on the day. Resolvers are still holding the old value for as long as the old TTL said.

Route 53 Resolver

- Every VPC has a resolver at the **base of the CIDR plus two**, and at the link-local address 169.254.169.253.
- By default it resolves **public names**, **private hosted zones** associated with the VPC, and **EC2 private DNS names**.
- It does **not** know about your on-premises domains, and on-premises systems cannot reach it directly.
- That gap is what **inbound and outbound endpoints** exist to close.

> ON THE EXAM
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> Module 29's flow log exclusion applies here: queries to the VPC resolver are never recorded in flow logs. Use Resolver query logging.

Inbound or outbound endpoint?

INBOUND ENDPOINT

On-premises asks AWS

- ENIs with private IPs in your subnets
- On-premises DNS forwards AWS domains to them
- Data centre systems resolve private zones
- Queries flow into the VPC

OUTBOUND ENDPOINT

AWS asks on-premises

- ENIs used as a source for outbound queries
- Paired with rules naming the domains
- Instances resolve corp.example.com
- Queries flow out of the VPC

THE TIEBREAKER

Name the direction of the query, not the traffic. Inbound means queries coming in to AWS; outbound means queries going out to your data centre.

Resolver rules

- A **forwarding rule** says which domain names go to which target DNS servers, through an outbound endpoint.
- A **system rule** overrides forwarding for a subdomain, so example.com forwards on-premises while aws.example.com stays with Route 53.
- Rules are **shareable through AWS RAM**, so a networking account defines them once for the whole organisation.
- The most specific matching rule wins, exactly as with route tables.

> ON THE EXAM
>
> The RAM sharing point matters at scale. Without it, every VPC in every account needs its own rules and outbound endpoints, and they drift.

Route 53 Profiles

- A **Profile** bundles **private hosted zones, Resolver rules and DNS Firewall rule groups** into one shareable object.
- Share it through **AWS RAM** and associate it with VPCs across accounts, so a new VPC inherits the whole DNS configuration at once.
- Without it, each element is associated separately with each VPC, which is where multi-account estates drift apart.
- Changing the Profile changes every associated VPC, which is the point and also the blast radius.

> ON THE EXAM
>
> The same pattern as StackSets and organization backup policies: define once centrally, and new resources inherit it without anyone remembering.

Resolver DNS Firewall

- Filters **outbound DNS queries from your VPCs** against domain lists, before the name is resolved.
- Actions are **ALLOW**, **BLOCK** or **ALERT**. A block can return NODATA, NXDOMAIN, or an override to a domain you choose.
- **AWS Managed Domain Lists** cover malware, botnet command and control, phishing and DGA domains, and are **free to use**.
- Rule groups are evaluated in **priority order**, and are associated with VPCs. AWS's guidance is to **test with ALERT before switching to BLOCK**.

> ON THE EXAM
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> Blocking at the DNS layer stops the connection before it is attempted, so no data leaves. Network Firewall inspects packets after a name has resolved.

Advanced DNS threat detection

- Beyond domain lists, DNS Firewall detects **DNS tunnelling**, where data is exfiltrated inside DNS queries themselves.
- That matters because DNS is often the one protocol allowed out of a restricted network, so it is the exfiltration path attackers reach for.
- It also detects **domain generation algorithm** patterns, where malware computes thousands of candidate domains rather than using a fixed one.
- A static domain list cannot catch either, because neither uses domains anybody has listed.

> ON THE EXAM
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> This is the DNS equivalent of GuardDuty's shift from signatures to behaviour: detecting the shape of an attack rather than a known name.

New: Route 53 Global Resolver

- Generally available **9 March 2026**, across 30 Regions, for IPv4 and IPv6. An **internet-reachable anycast resolver**.
- Resolves **public domains and your private hosted zones** from anywhere, through one set of anycast addresses.
- Clients are authorised by **Access Source**, matching an IP range, or by revocable **Access Tokens** for clients with changing addresses.
- Supports **Do53, DNS over HTTPS and DNS over TLS**, and carries the same DNS Firewall filtering.

> ON THE EXAM
>
> This removes the need for inbound endpoints plus split-DNS forwarding for branch offices and remote workers. Inbound endpoints remain right for a data centre with its own DNS servers.

The remaining pieces

- **Resolver query logging** records the queries your VPCs make. Flow logs never capture these, as module 29 established.
- **DNSSEC signing** for public hosted zones proves responses have not been tampered with. The key-signing key lives in **KMS in us-east-1**.
- DNSSEC is a **one-way door operationally**: an expired or mismatched key makes the domain unresolvable, not merely unverified.
- **Domain registration** covers 250 or so top-level domains, and creates the hosted zone for you.

> ON THE EXAM
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> Query logging is the only way to see what your instances are looking up. That is both a security signal and the fastest way to find a hard-coded endpoint.

> A team must point example.com, the bare domain with no subdomain, at an Application Load Balancer. They create a CNAME record for example.com pointing at the load balancer's DNS name, and Route 53 rejects it.
>
> A CNAME for www.example.com pointing at the same load balancer works correctly.

What should they do?

A. Create an alias record for example.com targeting the load balancer

B. Create an A record with the load balancer's IP addresses

C. Delete the SOA record so the CNAME can be created

D. Use a redirect from example.com to www.example.com in the application

◆ PAUSE AND DECIDE

What should they do?

A Alias record at the apex

Alias is Route 53's extension that works at the zone apex, follows the load balancer's changing addresses, and is free for AWS targets.

B A record with the IPs

A load balancer's addresses change without notice. This works until AWS replaces a node, then fails silently.

C Delete the SOA record

Would break the zone entirely. The DNS standard forbids a CNAME coexisting with other records at the apex, and SOA is mandatory.

D Application redirect

Requires something already answering at the apex to issue the redirect, which is the problem being solved.

THE SIGNAL

The comparison is supplied: www works, the apex does not. That is the CNAME-at-apex rule, and only an alias solves it.

> A team creates a private hosted zone for example.com and associates it with a VPC. Instances can resolve app.example.com internally, as intended. However, they can no longer resolve mail.example.com, a public record not defined in the private zone.
>
> External systems resolve mail.example.com normally.

What is the cause and the fix?

A The private hosted zone is authoritative for the whole domain inside the VPC. Add the missing records to it

B enableDnsHostnames is disabled on the VPC

C The private hosted zone needs a forwarding rule to the public zone

D The VPC needs an outbound Resolver endpoint

◆ PAUSE AND DECIDE

What is the cause and the fix?

A The private zone is authoritative for the whole domain

Inside an associated VPC the private zone answers for example.com entirely. Names not defined in it do not resolve, even if they exist publicly.

B enableDnsHostnames disabled

Would prevent the private zone working at all. app.example.com resolves correctly, so the attribute is enabled.

C Forward to the public zone

Resolver forwarding rules send queries to DNS servers you specify. There is no mechanism to fall through from a private zone to a public one.

D Outbound endpoint

Sends queries to on-premises DNS servers. The missing record is on the public internet, which the resolver already reaches.

THE SIGNAL

Overlaying a private zone on a public domain is a decision to own every name in it. There is no fall-through, and that surprises people.

> A hybrid environment needs two things. Instances in AWS must resolve names in corp.internal, which is served by DNS servers in the data centre. Servers in the data centre must resolve names in a Route 53 private hosted zone used by AWS workloads.
>
> The team has a Direct Connect connection in place.

What should the team configure?

A. An outbound endpoint with a forwarding rule for corp.internal, and an inbound endpoint for the data centre's DNS to forward to

B. An inbound endpoint only, with a forwarding rule for corp.internal

C. An outbound endpoint only, associated with both networks

D. Associate the private hosted zone with the on-premises network

◆ PAUSE AND DECIDE

What should the team configure?

A Both endpoints

Two directions of query means two endpoints. Outbound with a rule sends AWS queries to the data centre; inbound receives queries coming the other way.

B Inbound only

Handles queries arriving from the data centre. AWS instances would still have no way to reach corp.internal.

C Outbound only

Handles AWS querying on-premises. The data centre still has no AWS address to send its queries to.

D Associate the PHZ on-premises

Private hosted zones associate with VPCs only. There is no way to associate one with a network outside AWS.

THE SIGNAL

Two requirements in opposite directions means two endpoints. Options B and C each solve exactly half, which is what makes them plausible.

> A security team must stop workloads in 45 VPCs from resolving known malware and command-and-control domains, and must detect data exfiltration hidden inside DNS queries. They want the same policy applied automatically to VPCs created later.
>
> They do not want to maintain domain lists by hand.

What should they implement?

A DNS Firewall rule groups using AWS Managed Domain Lists and advanced threat detection, bundled into a Route 53 Profile shared through RAM

B A custom domain list of known bad domains, associated with each VPC

C AWS Network Firewall in each VPC

D Security group rules denying outbound port 53

◆ PAUSE AND DECIDE

What should they implement?

A Managed lists plus advanced detection, in a Profile

Managed lists remove the maintenance, advanced detection catches tunnelling that no list contains, and a shared Profile covers new VPCs.

B Custom lists per VPC

Exactly the hand-maintained lists ruled out, associated 45 times, and a list can never catch tunnelling or generated domains.

C Network Firewall

Inspects packets after a name has resolved. Useful alongside, and it does not stop the lookup or see inside DNS queries the same way.

D Block outbound port 53

Breaks all DNS resolution, since instances query the VPC resolver. It would take every workload offline.

THE SIGNAL

Three requirements: no manual lists, detect tunnelling, cover future VPCs. Managed lists, advanced detection, and a Profile map onto one each.

A company has 90 branch offices and a large remote workforce. All of them need to resolve names in Route 53 private hosted zones. The current design forwards from a DNS server at each site to Resolver inbound endpoints in two Regions, and the team maintains failover configuration at every site.

The current design forwards from a DNS server at each site to Resolver inbound endpoints in two Regions. The team maintains failover configuration at every site. Remote workers have changing IP addresses and cannot be covered by source-IP rules.

What should the company consider?

A Route 53 Global Resolver, using Access Sources for the offices and Access Tokens for remote workers

B Inbound Resolver endpoints in more Regions

C A Route 53 Profile shared to each branch office

D Make the private hosted zones public

◆ PAUSE AND DECIDE

What should the company consider?

A Global Resolver with both authorisation methods

Anycast addresses remove per-site failover configuration, Access Sources cover fixed-IP offices, and Access Tokens cover clients whose addresses change.

B More inbound endpoints

More of the thing causing the burden. Every site still needs forwarding and failover configuration maintained by hand.

C A Profile per office

Profiles associate with VPCs. There is no way to associate one with a branch office network.

D Make the zones public

Publishes internal DNS records to the entire internet, which is a serious disclosure of internal architecture.

THE SIGNAL

The remote-worker clause is decisive: changing IPs cannot be authorised by source range, and Access Tokens exist precisely for that case.

The five things to carry forward

- **A CNAME cannot exist at the zone apex.** That is a DNS rule, and an alias record is the answer.
- **A private hosted zone is authoritative for the whole domain** inside associated VPCs. There is no fall-through.
- **Inbound means queries coming in to AWS; outbound means queries going out.** Most hybrid designs need both.
- **DNS Firewall blocks before resolution**, and detects tunnelling and generated domains that no list contains.
- **Global Resolver (March 2026)** gives anycast resolution of private zones from anywhere, without inbound endpoints.

> ON THE EXAM
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> Next: CloudFront and the edge - OAC, cache behaviours, invalidation, WAF, Shield and Global Accelerator.